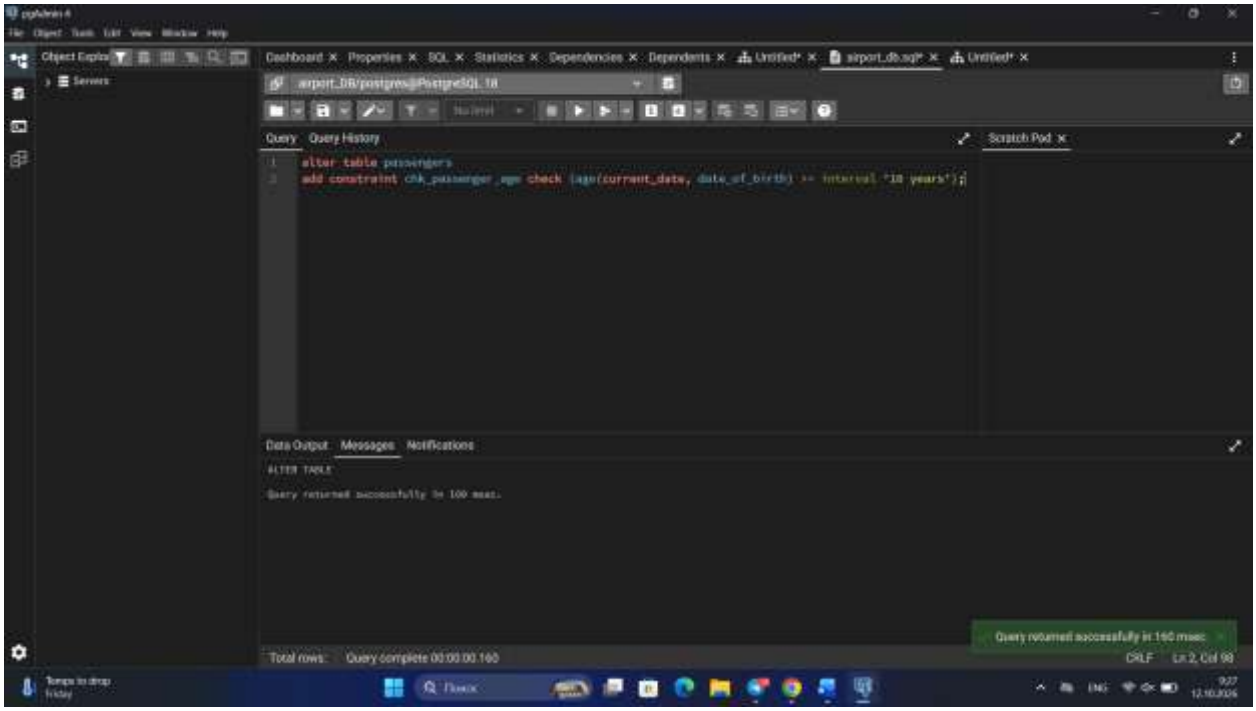
Laboratory work 5

We continue to work with the database from the previous laboratory works.

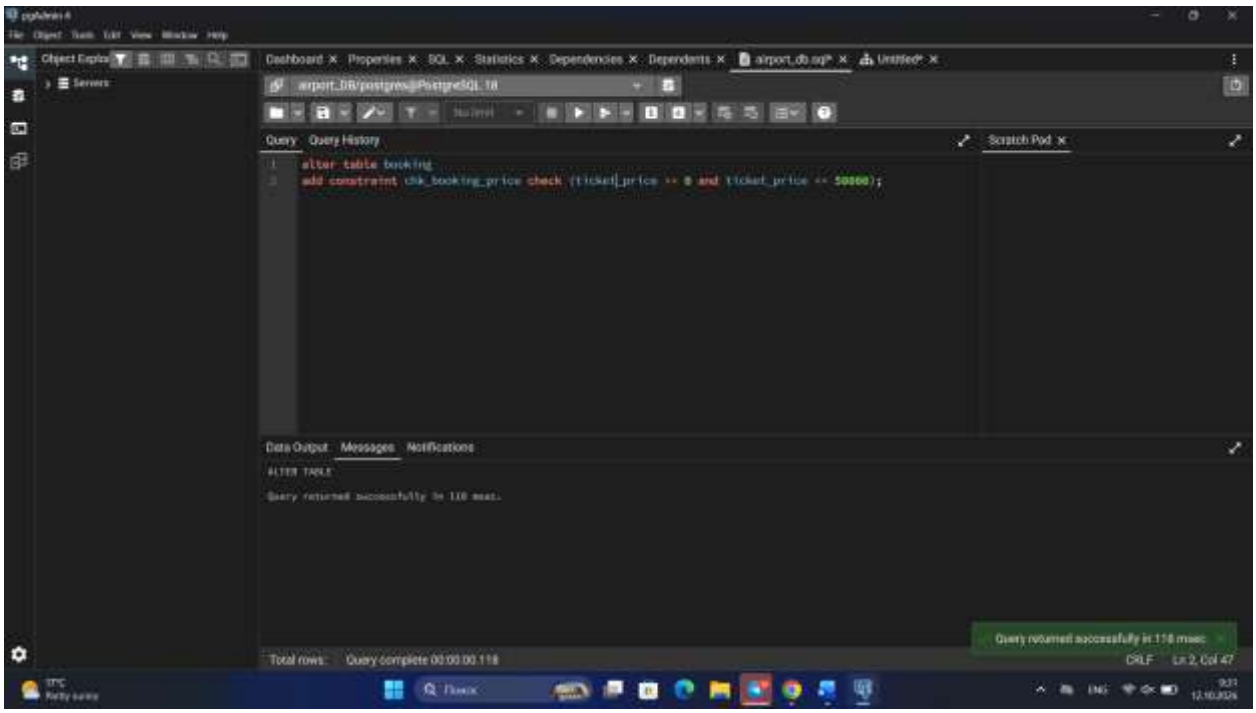
Take a full-page screenshot that covers the code and results of each task.

Tasks:

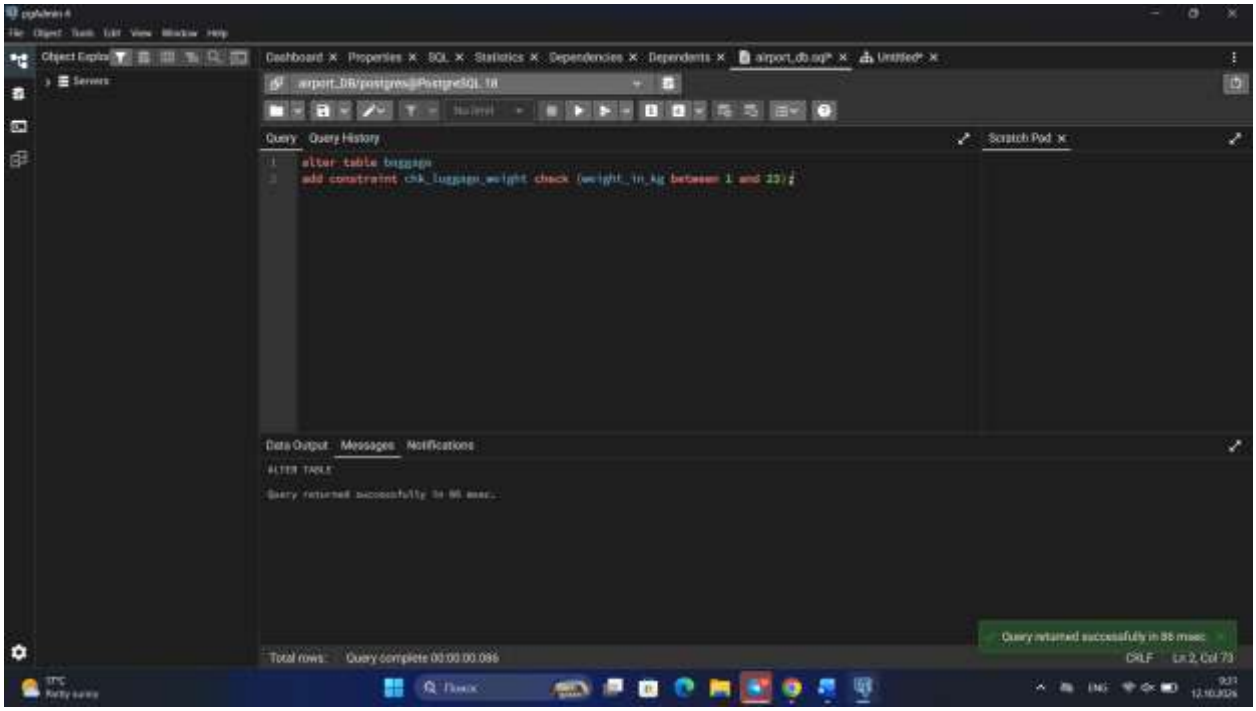
1. Add a CHECK constraint to passenger table to provide that passengers must be at least 10 years old.



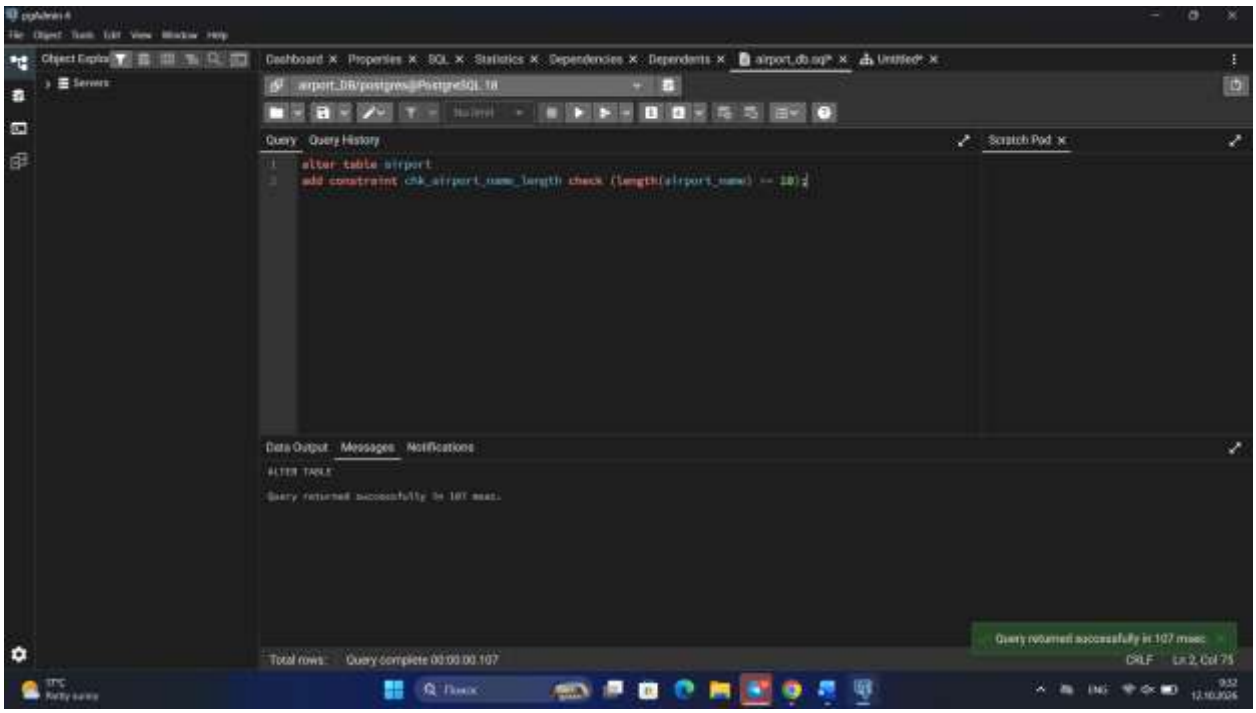
2. Add a CHECK constraint to accept values in booking price not more than 50000tg and less than 0tg.



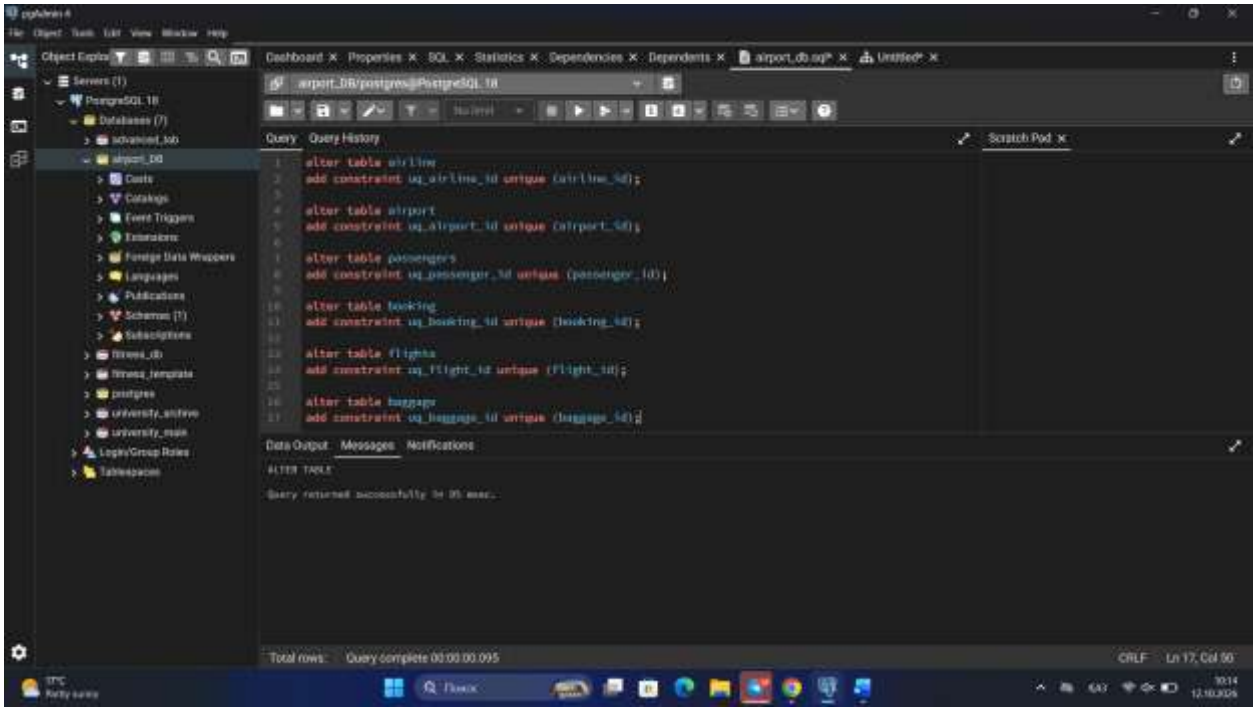
3. Add a CHECK constraint to accept the luggage weight between 1 and 23 kg.



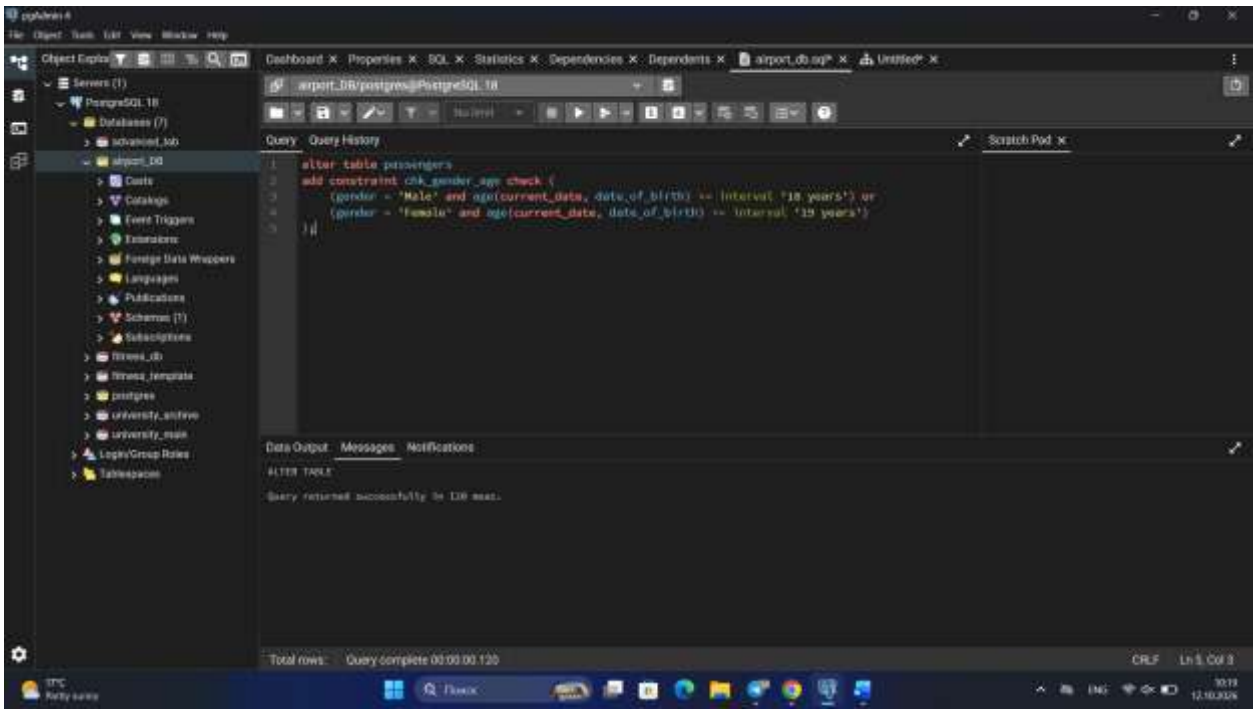
4. Add a CHECK constraint to ensure that all values in airport_name must have at least 10 characters.



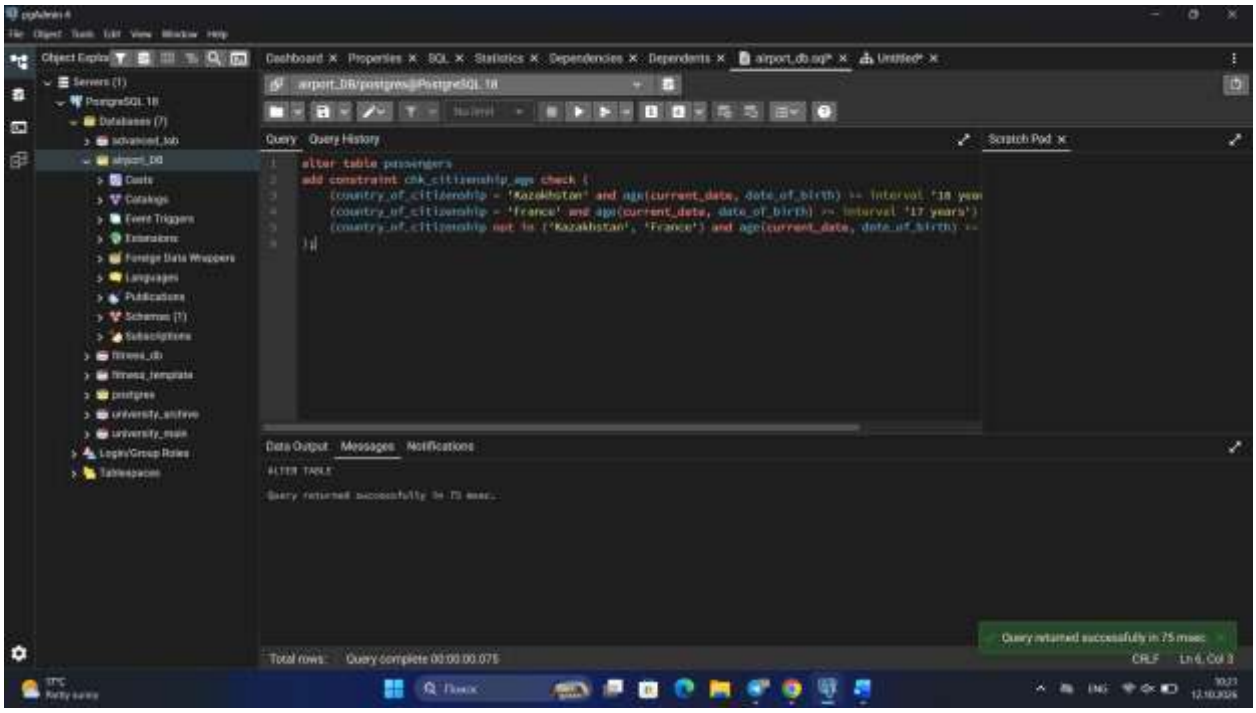
5. Add UNIQUE constraint to some columns in each table in database.



6. Add a CHECK constraint to ensure that male passengers must be at least 18 years old and female passengers must be 19 years old.

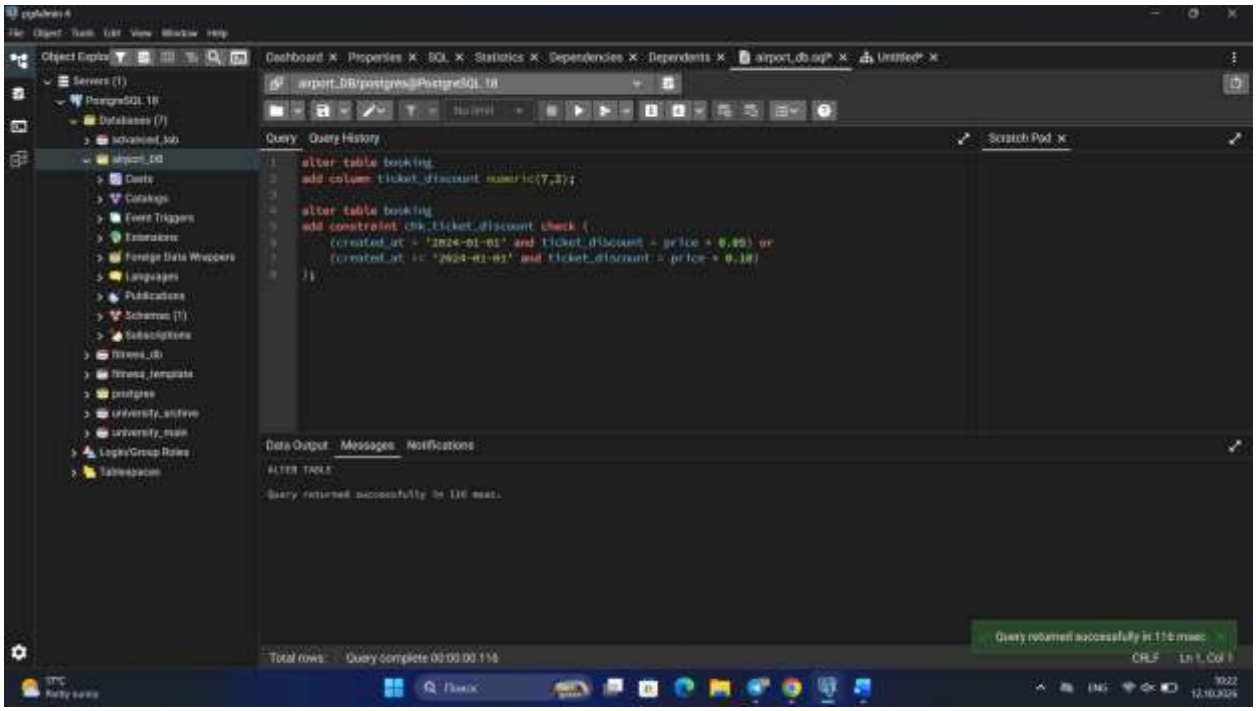


7. Add a CHECK constraint to add rule as follow (use column country_of_citizenship):
    - Passengers from Kazakhstan must be at least 18 years old.
    - Passengers from France must be at least 17 years old.
    - Passengers from other countries must be at least 19 years old.

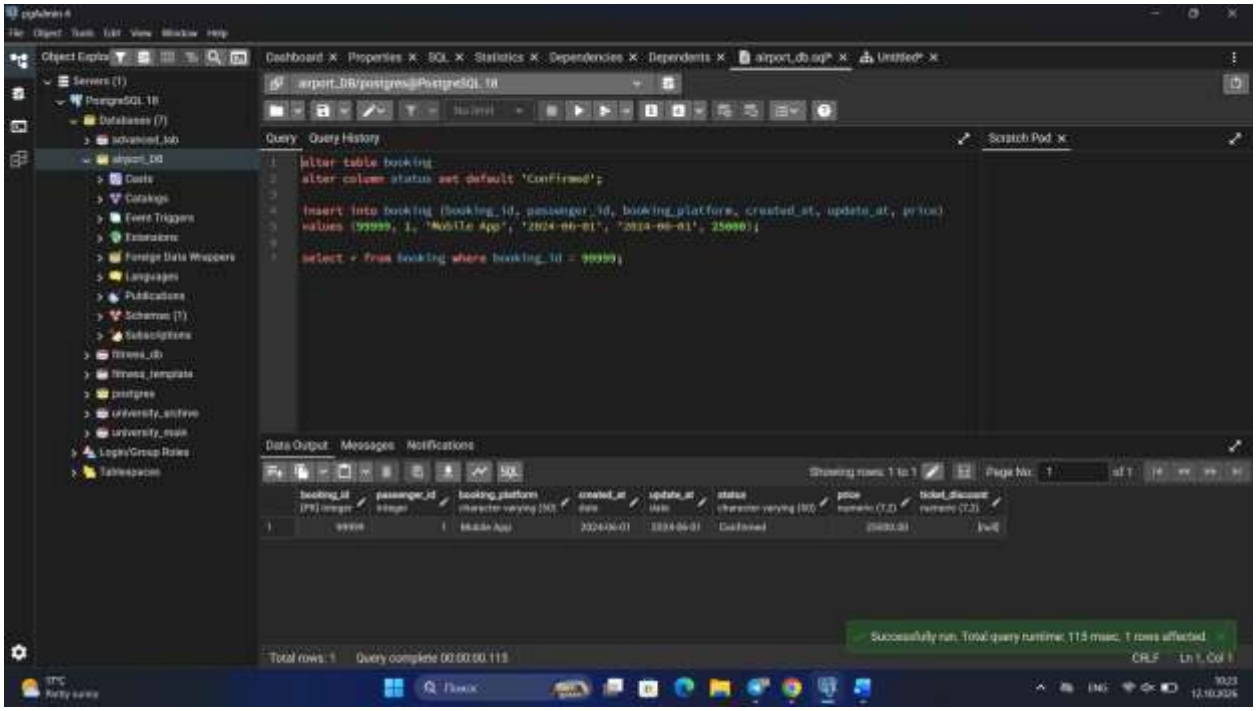


8. Add a ticket_discount column to table booking and a CHECK constraint to apply some discount based on ticket price and created time:
    - the constraint applies a 5% discount for tickets created after 2024-01-01,

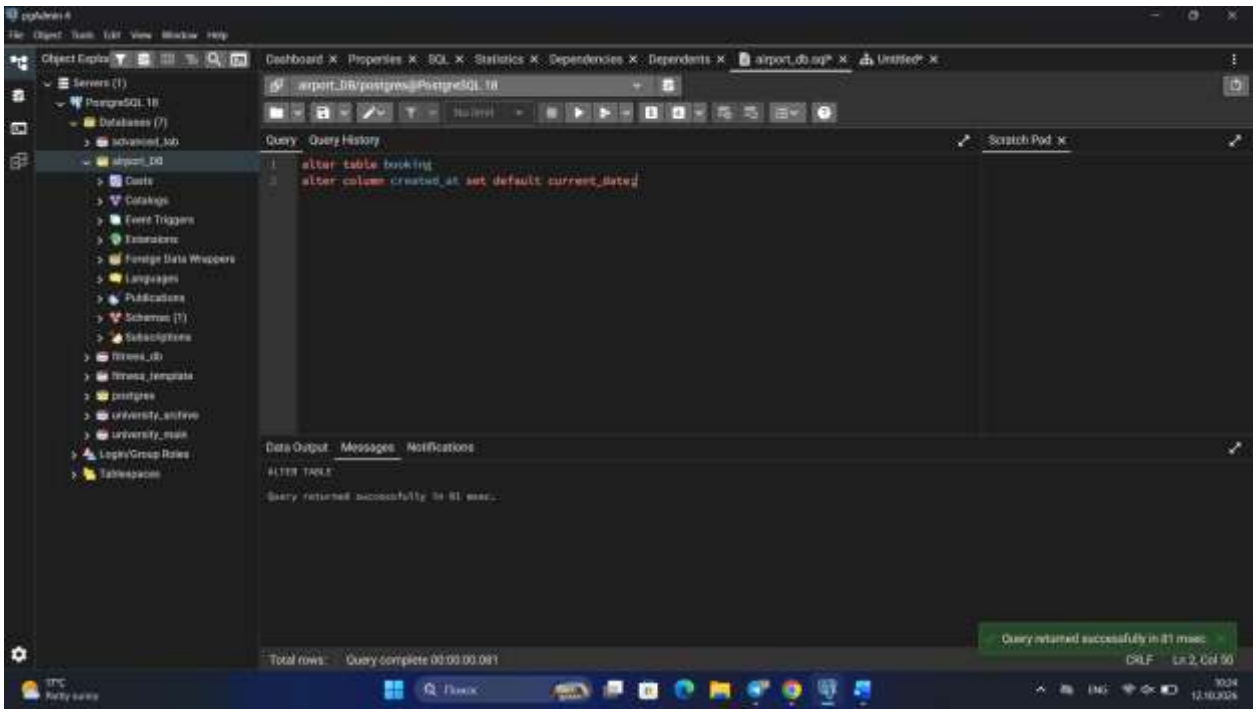
and 10% discount for tickets created before 2024-01-01.



9. Modify the booking table so that the booking status column has a default value of 'Confirmed'. Then insert a new booking without specifying booking status.



1. Modify the appropriate date column in the database so that when a new record is inserted, the date is automatically set to the current date.



Conclusion: In this laboratory work, we successfully practiced advanced DDL operations in PostgreSQL using ALTER TABLE to modify database schemas. We implemented various CHECK constraints to enforce business rules (such as age restrictions based on gender and citizenship, price limits, luggage weight ranges, and string length validation), added UNIQUE constraints to maintain data integrity across tables, and configured default column values for automatic tracking such as dates and booking statuses.